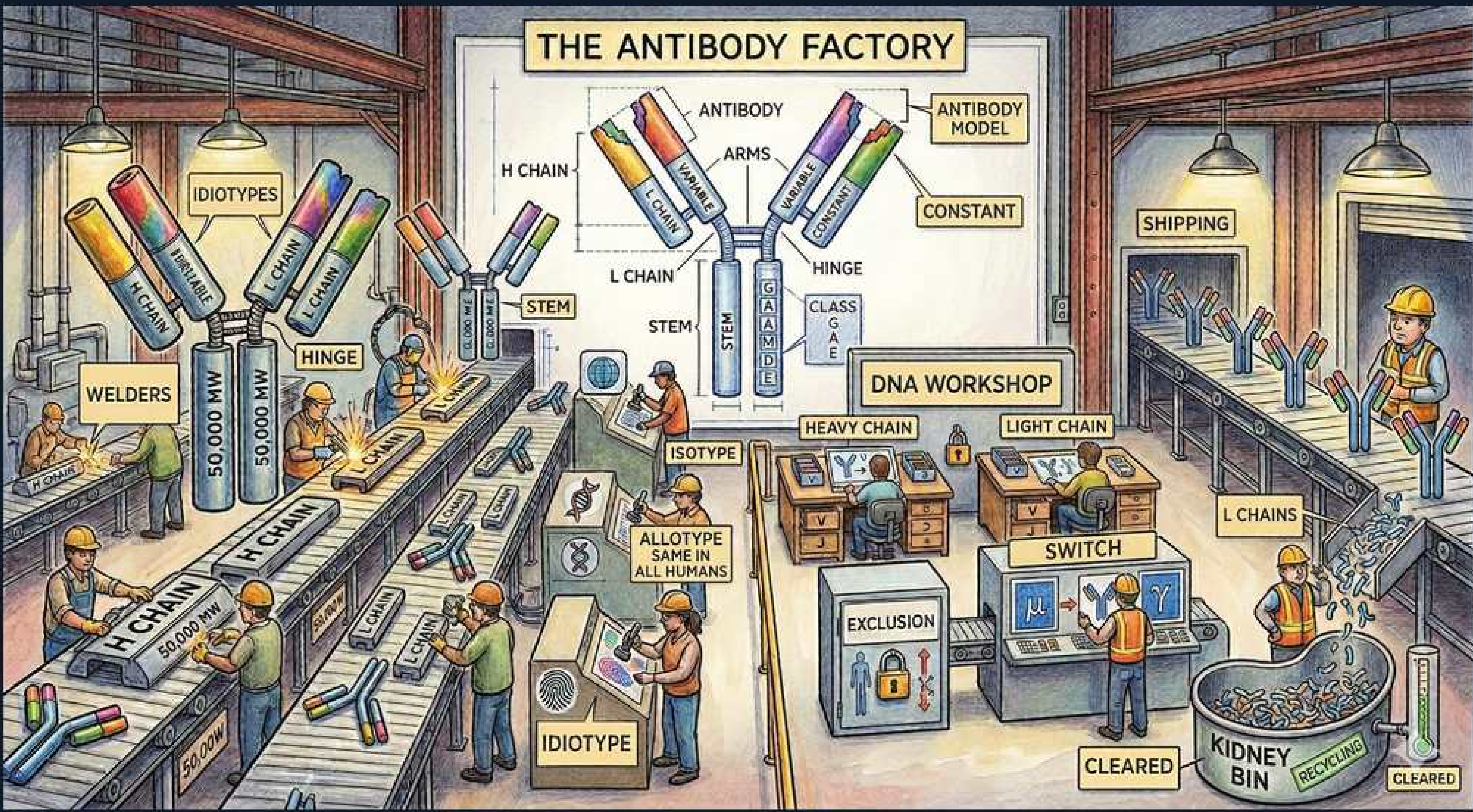
Plasma Cell — The Antibody Factory (Immunoglobulin Structure)



1. Antibody model (Y shape)

WHAT YOU SEE

The big labelled ANTIBODY MODEL on the wall — a Y with two ARMS and a central STEM — the single identical Y is the monoclonal (M) protein churned out by one plasma-cell clone.

WHAT IT ENCODES

Each immunoglobulin monomer = 2 identical heavy chains + 2 identical light chains joined by interchain disulfide bonds, forming the Y. The two Fab arms bind antigen; the Fc stem mediates effector function. In multiple myeloma a single plasma-cell clone overproduces ONE identical Y = the monoclonal (M) protein, seen as a sharp narrow spike in the gamma region on SPEP and confirmed by immunofixation. Diagnostic CRAB myeloma is defined by ≥10% clonal marrow plasma cells (or plasmacytoma) PLUS end-organ damage.

BOARD TRAP

WRONG to call any monoclonal spike 'myeloma.' MGUS = M-protein <3 g/dL, marrow plasma cells <10%, and NO CRAB/SLiM criteria; smoldering myeloma = M-protein ≥3 g/dL or 10–60% plasma cells but still no end-organ damage. Discriminator: it is end-organ damage (CRAB) or a SLiM biomarker (≥60% plasma cells, FLC ratio ≥100, >1 focal MRI lesion) that upgrades to treatment-requiring myeloma, not the spike alone.

2. Heavy chain (H chain)

WHAT YOU SEE

H CHAIN tags running down the two long inner backbone struts of the Y — the heavy-chain constant region is what stamps the antibody's class (IgG, IgA, etc.).

WHAT IT ENCODES

The heavy-chain constant region defines the isotype (class) of the antibody. IgG (~50–60% of myeloma) is most common, then IgA (~20%); IgD and IgE myeloma are rare and aggressive. Free light-chain-only ('light-chain' or Bence-Jones) myeloma makes NO intact heavy chain (~15–20%). An IgM monoclonal points away from classic myeloma toward Waldenström. Heavy chain mnemonic for classes: 'GAMDE' (G, A, M, D, E).

BOARD TRAP

WRONG to diagnose multiple myeloma when the M-protein is IgM. An IgM monoclonal = Waldenström macroglobulinemia (lymphoplasmacytic lymphoma, MYD88 L265P mutation) and presents with hyperviscosity, not lytic bone lesions. Discriminator: IgM + lymphoplasmacytic marrow = Waldenström; IgG/IgA + lytic bone/CRAB = myeloma.

3. Light chain (L chain)

WHAT YOU SEE

L CHAIN tags on the two shorter outer struts capping the Y's arms — each antibody carries only ONE light-chain flavour, kappa or lambda, never both.

WHAT IT ENCODES

Every antibody uses ONE light-chain type — kappa (κ) or lambda (λ) — never both; normal serum κ:λ ratio is ~2:1 (free light-chain assay normal ratio 0.26–1.65). Malignant clones overproduce one type, skewing the ratio dramatically (involved/uninvolved). Serum free light chains are key for diagnosis, the SLiM criterion (FLC ratio ≥100), and tracking light-chain-only and oligosecretory myeloma where SPEP may be unrevealing.

BOARD TRAP

WRONG to rely on SPEP alone to exclude myeloma. Light-chain-only myeloma often shows little/no SPEP spike because free light chains are rapidly filtered by the kidney. Discriminator: order serum free light-chain assay + urine immunofixation (UPEP) — a markedly abnormal κ/λ ratio reveals the clone SPEP misses.

4. Variable / idiotype region

WHAT YOU SEE

IDIOTYPES stamped at the very fingertips of the Y's arms on the assembly line — the unique antigen-grip at the tips is the clone-specific idiotype, the literal meaning of 'monoclonal'.

WHAT IT ENCODES

The variable (V) region at the Fab tips contains the hypervariable CDRs that form the unique antigen-binding site = the idiotype. A single malignant clone makes ONE identical idiotype, which is the molecular basis of 'monoclonal.' Diversity here is generated by VDJ recombination plus somatic hypermutation in germinal centers. The idiotype is patient-specific and tumor-specific (basis for some personalized vaccine/immunotherapy strategies).

BOARD TRAP

WRONG to equate idiotype with allotype or isotype. Idiotype = the antigen-binding variable region unique to one clone; isotype = the class (IgG/A/M/D/E) set by the heavy-chain constant region; allotype = inherited constant-region polymorphisms differing between individuals. Discriminator: only the idiotype is clone-specific and defines monoclonality.

5. Constant region

WHAT YOU SEE

CONSTANT label on the Y's stem (the Fc base) — the constant stem is the business end that fixes complement and binds Fc receptors.

WHAT IT ENCODES

The constant (Fc) region of the heavy chain determines isotype class and effector functions: complement fixation, Fc-receptor binding, opsonization, and (for IgG) placental transfer. Class switching alters only this heavy-chain constant region while leaving the variable region/antigen specificity intact. Therapeutic anti-CD38 (daratumumab) and SLAMF7 (elotuzumab) antibodies exploit Fc-mediated ADCC/CDC against plasma cells.

BOARD TRAP

WRONG to think class switching changes what antigen the antibody binds. Class switch recombination swaps the constant region (IgM→IgG/IgA/IgE) but the variable region/idiotype/antigen specificity is preserved. Discriminator: effector function changes, target antigen does not.

6. Hinge

WHAT YOU SEE

HINGE label at the fork where the Y's arms meet the stem — this flexible joint lets the arms swing to grab two antigens and is where papain vs pepsin cut.

WHAT IT ENCODES

The hinge region, rich in proline and disulfide bonds, gives the Fab arms flexibility to bind two antigen epitopes at variable angles and exposes the Fc for effector engagement. It is the site cleaved by papain (yields 2 Fab + 1 Fc) versus pepsin (yields a single F(ab')2 + degraded Fc). IgG and IgA have prominent hinges; IgM and IgE lack a true hinge (use an extra CH domain instead).

BOARD TRAP

WRONG to confuse papain and pepsin cleavage. Papain cuts ABOVE the hinge disulfides → 2 monovalent Fab + 1 Fc. Pepsin cuts BELOW → one bivalent F(ab')2 (still cross-links antigen) + digested Fc. Discriminator: pepsin preserves bivalent binding (F(ab')2 can still agglutinate); papain does not.

7. Class switch station

WHAT YOU SEE

SWITCH workbench in the DNA WORKSHOP where workers swap out the heavy-chain base — class switching changes only the constant region (IgM to IgG/IgA/IgE), keeping the same antigen grip.

WHAT IT ENCODES

Class switch recombination (CSR) occurs in germinal-center B cells, directed by T-cell help (CD40L–CD40) and cytokines, and requires the enzyme AID (activation-induced cytidine deaminase). It changes the heavy-chain isotype from IgM/IgD to IgG, IgA, or IgE while keeping the same antigen specificity. AID also drives somatic hypermutation for affinity maturation.

BOARD TRAP

WRONG to attribute failed class switching to a B-cell-intrinsic antibody problem only. Hyper-IgM syndrome (high IgM, low IgG/IgA/IgE, recurrent sinopulmonary infections + Pneumocystis/Cryptosporidium) is most often X-linked from a CD40L defect on T cells — a T-cell help failure. Discriminator: AID or CD40/CD40L defect blocks CSR; X-linked CD40L deficiency is the classic boards answer.

---

8. Allelic exclusion

WHAT YOU SEE

A chained PADLOCK box marked EXCLUSION shutting down the second allele — allelic exclusion locks each cell to ONE heavy chain and ONE light-chain type, the basis of clonality.

WHAT IT ENCODES

Allelic exclusion ensures each B cell/plasma cell productively rearranges and expresses only ONE heavy-chain allele and ONE light-chain type (κ OR λ). This guarantees a single antigen specificity per cell and is the developmental basis of clonality — and therefore of the monoclonal protein in myeloma. A successful rearrangement signals to shut down the other allele.

BOARD TRAP

WRONG to expect a malignant plasma-cell clone to make both κ and λ. Allelic (and isotype) exclusion means a true monoclonal proliferation is restricted to a single light chain — demonstrating κ OR λ restriction (e.g., on flow/IHC) confirms clonality versus a reactive polyclonal (mixed κ and λ) plasmacytosis. Discriminator: light-chain restriction = neoplastic; polyclonal mix = reactive.

---

9. Allotype

WHAT YOU SEE

The ALLOTYPE — SAME IN ALL HUMANS sign — allotypes are inherited constant-region variants that differ between people, not antigen-driven like the idiotype.

WHAT IT ENCODES

Allotypes are inherited polymorphic variants of the constant region that differ between individuals within a species (e.g., Gm and Km markers). They are germline-encoded, NOT antigen-driven. Clinically relevant because allotype mismatch can provoke anti-allotype responses (e.g., after transfusion/IVIG or against therapeutic antibodies).

BOARD TRAP

WRONG to equate allotype with idiotype. Allotype = inherited constant-region differences BETWEEN people (same across a person's antibodies); idiotype = variable-region antigen-binding differences unique to a clone. Discriminator: allotype is germline/constant-region, idiotype is somatically generated/variable-region.

---

10. Free light chains to kidney

WHAT YOU SEE

The L CHAINS conveyor belt dumping loose light chains straight into the KIDNEY BIN — excess free light chains flood the kidney and precipitate as obstructive casts (cast nephropathy).

WHAT IT ENCODES

Excess free light chains are filtered at the glomerulus and reabsorbed in the proximal tubule; overload precipitates them with Tamm-Horsfall (uromodulin) protein in the distal tubule, forming obstructive casts = cast nephropathy (myeloma kidney), the leading cause of myeloma renal failure (the 'R' of CRAB). Management: aggressive hydration, treat the clone fast (bortezomib-based, no renal dose adjustment), avoid nephrotoxins/contrast/NSAIDs.

BOARD TRAP

WRONG to exclude myeloma kidney because the urine dipstick is negative for protein. Standard dipstick detects ALBUMIN, not light chains — so Bence-Jones proteinuria reads negative. Discriminator: order UPEP with immunofixation (or serum free light chains); a negative dipstick with heavy proteinuria on 24-hr collection or sulfosalicylic acid test signals light chains.

---

11. Isotype

WHAT YOU SEE

ISOTYPE tag beside the heavy-chain assembly station — isotype is the antibody class set by the heavy-chain constant region (named by immunofixation, e.g. 'IgG kappa').

WHAT IT ENCODES

Isotype = the antibody class (IgG, IgA, IgM, IgD, IgE) defined by the heavy-chain constant region; the M-protein's isotype is determined by immunofixation electrophoresis. IgG is the most common myeloma isotype and the most abundant serum Ig; IgA myeloma can co-migrate in the beta region and be missed on SPEP. Knowing the isotype guides expectations (e.g., IgA/IgM → hyperviscosity risk).

BOARD TRAP

WRONG to read SPEP alone as the M-protein characterization. SPEP only shows a spike; immunofixation (IFE) is required to assign the heavy-chain isotype AND light-chain type (e.g., 'IgG kappa'). Discriminator: SPEP quantifies, IFE identifies — IFE is also the test that confirms deep treatment responses (IFE-negative).

---

12. DNA workshop (gene assembly)

WHAT YOU SEE

The DNA WORKSHOP banner over the gene-assembly workbenches — this is where the antibody genes are built: VDJ recombination in marrow, then hypermutation/class-switch in germinal centers.

WHAT IT ENCODES

Antibody diversity is generated in two stages: (1) VDJ recombination via RAG1/RAG2 in the bone marrow (combinatorial + junctional diversity) creates the naive repertoire; (2) somatic hypermutation + class switching via AID in germinal centers refine affinity and class after antigen exposure. Defects illustrate the pathway: RAG mutations → SCID (no B/T cells); AID/UNG mutations → hyper-IgM (autosomal recessive form).

BOARD TRAP

WRONG to attribute all combinatorial diversity to somatic hypermutation. VDJ recombination (RAG1/RAG2, in marrow, antigen-INDEPENDENT) builds the initial repertoire; somatic hypermutation (AID, in germinal centers, antigen-DEPENDENT) fine-tunes affinity afterward. Discriminator: RAG/marrow/before antigen vs AID/germinal center/after antigen.
